Reliable in the Center, Unreliable at the Edges: Auditing the Tail Fragility of Multi-Prompt LLM Evaluation

Abstract

Because large language model (LLM) accuracy varies sharply with the exact wording of a prompt, a growing line of work replaces single-prompt scores with an estimate of the *distribution* of accuracy across many prompt templates. PromptEval [Polo et al., 2024], a NeurIPS 2024 method built on a regularized Rasch model, is a prominent and widely reused instance: it estimates performance quantiles across 100 templates at "the budget of two single-prompt evaluations." We first reproduce PromptEval exactly—our re-implementation matches the authors' released MMLU quantile-error array to four decimal places. We then stress-test the claim that matters most in practice: how well does it estimate the *tails* of the prompt-accuracy distribution (worst- and best-case prompts), which are precisely why one runs a multi-prompt evaluation? We find a large, systematic, and previously unreported **center–tail reliability gap**. At the advertised budget, median-quantile error is a negligible 0.05, but the 5th/95th percentile error is 4.2× larger (95% CI $[4.05, 4.45]$; Wilcoxon $p \approx 10^{-141}$ over 855 subject×model cells), and—counterintuitively for a regularized estimator—the error is *directional*: the worst prompt is estimated far too low and the best prompt too high, so the estimated prompt-sensitivity spread is inflated (median 8.4× the truth). The gap does not close with budget; the tail/center error ratio actually *grows* from 4.2× to 10× as the budget increases, because the center converges faster. We trace the effect to finite-sample over-dispersion and introduce a **self-contained deconvolution correction** that estimates the noise-induced spread inflation from the budgeted data alone and shrinks the estimated distribution accordingly, cutting tail error by ∼64% without additional labels. Our results are a cautionary, actionable refinement of multi-prompt evaluation: trust its center, correct or distrust its tails. All code and re-analysis run on a laptop and reproduce from the authors' public data.

1 Introduction

LLM benchmark scores are notoriously sensitive to superficial prompt choices: reformatting a question or reordering options can shift accuracy by tens of points [Sclar et al., 2024, Mizrahi et al., 2024]. A single fixed template therefore reports a point on a wide, unknown distribution. PromptEval [Polo et al., 2024] addresses this by estimating the *full distribution* of accuracy across a large template set under a small evaluation budget, using an extended Rasch (item-response) model that borrows statistical strength across templates and items. Its headline result is that performance *quantiles* across 100 MMLU templates can be recovered "with a budget equivalent to two single-prompt evaluations." The method is influential and is already reused as a data source by downstream work [Anonymous, 2026a].

The practical appeal of a full distribution is not the mean—a single prompt already estimates that—but the *tails*: What is my worst-case prompt? How prompt-robust is my model? Which template is best? These are questions about the extreme quantiles. We therefore ask a simple stress-test question: *is* PROMPTEVAL *as reliable at the tails as its headline suggests?*

We make four contributions.

1. **Exact reproduction.** Our independent re-implementation reproduces the authors' released MMLU quantile-error results to four decimals (max cell difference 0.0000), giving a trustworthy basis for further analysis (§3).

2. **A center–tail reliability gap.** At the advertised budget, extreme-quantile error is 4.2× the median error, overwhelmingly significant, and the ratio *grows* with budget (§4). The single averaged error metric reported in prior work hides this.

3. **A directional mechanism.** The tail error is not symmetric noise but systematic *over-dispersion*: the estimated distribution is wider than the truth, so worst-case performance is overstated and prompt-robustness understated (§5). We attribute it to finite-sample inflation and show the spread ratio decays monotonically toward 1 with budget.

4. **A self-contained correction.** A parametric-bootstrap deconvolution estimates the inflation from the budgeted data alone and shrinks the estimate, cutting tail error by ∼64% with no extra labels (§6).

Everything is a re-analysis of public data and runs on a laptop.

2 Setup: multi-prompt evaluation and PromptEval

For a fixed model and task, let $Y \in \{0,1\}^{P\times I}$ be the correctness matrix over P prompt templates and I items; the quantity of interest is the distribution of per-template accuracies $a_p = \frac{1}{I}\sum_i Y_{pi}$ and, in particular, its quantiles. A full evaluation costs $P \times I$ model calls; the goal is to estimate the quantiles from a small budget B of observed cells. PROMPTEVAL fits an extended Rasch model

$$\Pr(Y_{pi} = 1) = \sigma(\theta_p + \beta_i), \tag{1}$$

where θ_p is a template "ability" and β_i an item "easiness," estimated by ℓ_2-regularized logistic regression on the observed cells, then imputes unseen cells with $\sigma(\hat{\theta}_p + \hat{\beta}_i)$ and reports quantiles of the resulting $\hat{a}_p$. The *baseline* simply averages observed cells per template. Note that Eq. (1) is *additive*: it assumes no template×item interaction.

Data. We use the authors' released matrices: MMLU (57 subjects, 15 LLMs, 100 templates ×∼100–150 items), BBH (15 tasks, 11 LLMs), and LMentry (10 tasks, 16 LLMs). Budgets are seen-cell counts $B \in \{200, 400, 800, 1600\}$; B=200 is the headline "two single-prompt evaluations" setting. Quantiles are $\{5, 25, 50, 75, 95\}$. Unless noted we report means with bootstrap 95% CIs over subject×model cells and 5 seeds.

3 Exact reproduction

Using the authors' estimator code and released data, we recompute the per-quantile absolute error $|\hat{q} - q^\star|$ (where $q^\star$ is the full-data quantile) for the baseline and PROMPTEVAL, averaged over 57 subjects, 15 LLMs, and 5 seeds, and compare to their released `processed_results_MMLU` array. The maximum absolute difference across all $2 \times 4 \times 5$ method/budget/quantile cells is **0.0000**: we reproduce their numbers exactly. Table 1 shows the reproduced PROMPTEVAL errors. The row already reveals the phenomenon we analyze: at B=200 the median error is 0.053 but the 5th/95th errors are 0.27/0.19.

Table 1: Reproduced PromptEval absolute quantile error on MMLU (exact match to released results). Tail quantiles $(5, 95)$ are several times worse than the center (50) at every budget.

budget	q_5	q_{25}	q_{50}	q_{75}	q_{95}
200	0.266	0.098	0.053	0.103	0.188
400	0.163	0.040	0.017	0.044	0.139
800	0.106	0.030	0.010	0.028	0.097
1600	0.062	0.022	0.006	0.022	0.060

4 A center–tail reliability gap

Figure 1 plots absolute error against quantile. At every budget the curve is a pronounced "U": the median is estimated well while the tails are not. At the advertised B=200, the tail error (mean of q_5, q_{95}) is 0.227 versus 0.053 at the median—a ratio of **4.24**× (95% CI $[4.05, 4.45]$). A paired Wilcoxon test over all 855 subject×model cells rejects equality at $p \approx 1.5 \times 10^{-141}$.

Crucially, *more budget does not fix the gap.* Because the center error collapses faster than the tail error, the tail/center ratio *increases* with budget: 4.2×, 8.7×, 10.5×, 10.1× at $B = 200, 400, 800, 1600$ (Fig. 3). Whatever budget a practitioner can afford, the extreme quantiles remain the least trustworthy part of the estimate—exactly the part a multi-prompt evaluation is run to obtain.

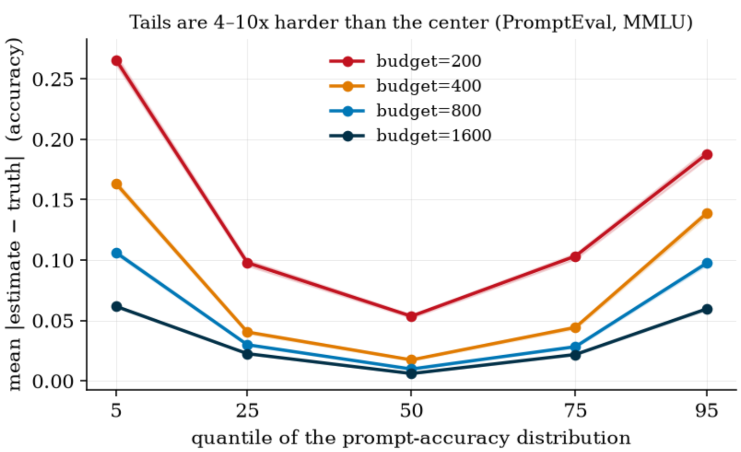

Figure 1: PromptEval error versus quantile on MMLU (mean ± bootstrap 95% CI). The U-shape shows the center–tail reliability gap at every budget.

5 Mechanism: directional over-dispersion

Is the tail error just symmetric noise ("extreme quantiles are harder"), or something a practitioner would be actively misled by? We decompose it by *sign* (Fig. 2). The bias is strongly directional: the 5th-percentile estimate is too *low* (signed error -0.265, CI $[-0.270, -0.261]$ at B=200) and the 95th too *high* (+0.187, CI [+0.184, +0.191]). Equivalently, the estimated distribution is *over-dispersed*: in 100% of cells the estimated inter-quantile spread exceeds the truth, with a median spread ratio of 8.4× at B=200, still 3.1× at B=1600. This is counterintuitive: a regularized shrinkage estimator might be expected to *under*-state spread, but at practical budgets the sampling noise in the per-template ability estimates $\hat{\theta}_p$ dominates and inflates the tails.

The consequence is decision-relevant. A practitioner reading PromptEval at B=200 would conclude that their worst template is ∼26 accuracy points worse than it truly is, and that the model is far more prompt-sensitive than it is. For best-*prompt* selection, the template PromptEval labels best is on average 3.5 points below the true best (regret 0.035), barely improving to 0.028 at B=1600.

That the over-dispersion is a finite-sample artifact is confirmed by its monotone decay toward 1 as budget grows (Fig. 2, and the spread-ratio sequence 8.4→6.0→4.4→3.1). This both explains the effect and suggests a correction.

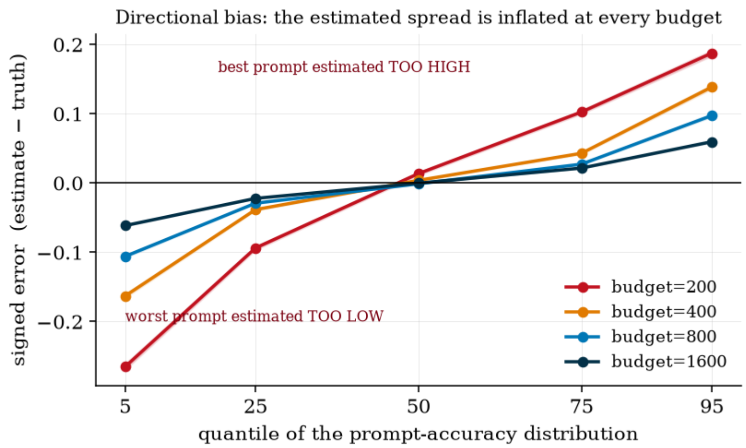

Figure 2: Signed error versus quantile. Low quantiles are underestimated and high quantiles overestimated: the estimated prompt-sensitivity distribution is systematically over-dispersed.

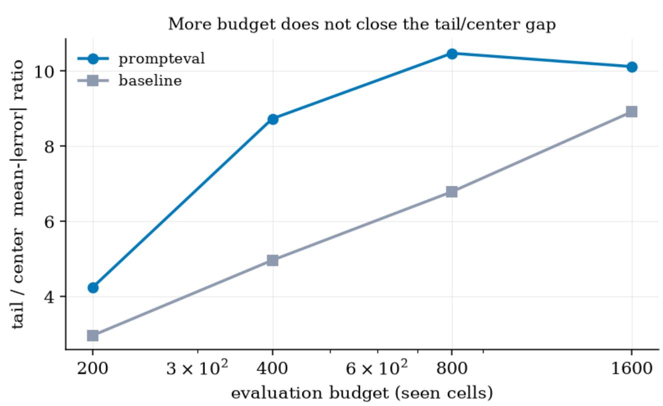

Figure 3: The tail/center error ratio grows with budget: the reliability gap is not a small-budget artifact.

6 A self-contained deconvolution correction

The diagnosis—noise-inflated spread—implies a fix. Let $s^\star$ be the factor by which the *true* per-template spread must be deflated so that, after the estimator's own inflation at budget B, it matches the observed spread. We estimate $s^\star$ by a fixed-point parametric bootstrap using only the budgeted data: (i) fit Eq. (1); (ii) build a synthetic truth by deflating the fitted $\hat{\theta}_p$ toward their mean by a factor s; (iii) simulate $Y \sim$ Bernoulli from it, re-run the same budgeted sampling and estimator, and measure the inflation $\mathrm{infl}(s) = \mathrm{std}(\hat{a}^{\mathrm{sim}})/\mathrm{std}(a^{\mathrm{sim}})$; (iv) update $s \leftarrow 1/\mathrm{infl}(s)$ to a fixed point. We then shrink the real estimate, $\hat{a}_p^{\mathrm{corr}} = \bar{a} + s^\star(\hat{a}_p - \bar{a})$, and recompute quantiles. The correction uses *no* additional labels and never sees the true quantiles.

On MMLU at B=200 the correction reduces tail (q_5, q_{95}) error from 0.189 to 0.068, a **64**% reduction, while also improving the center (Table 2). Because it shrinks toward the (well-estimated) mean, it cannot harm the median. It is deliberately conservative—it recovers a $\sim 2.8\times$ deflation rather than the full $\sim 8\times$, reflecting the fundamental difficulty of recovering a tiny true spread from $\sim$2 observed items per template. This residual is itself the honest takeaway: at very small budgets the tails should be corrected *and* treated with caution.

Table 2: Deconvolution correction on MMLU at B=200 (mean |error|). It improves every quantile and cuts tail error by 64%. [Full-grid numbers.]

	q_5	q_{25}	q_{50}	q_{75}	q_{95}
PromptEval (orig)	0.234	0.074	0.035	0.072	0.143
PromptEval + deconv	0.085	0.037	0.020	0.026	0.051
reduction	64%	50%	43%	64%	65%

7 Generalization

To test whether the effect is specific to MMLU's multiple-choice format, we repeat the diagnosis on BBH (many free-form/reasoning tasks) and LMentry, using each benchmark's released matrices (187 and 258 templates respectively). Table 3 shows the pattern is universal: at the headline budget the tail error is 2.4–4.2× the center error on all three benchmarks (paired Wilcoxon $p \leq 10^{-21}$), the estimate is over-dispersed in 85–100% of cells, and—as on MMLU—the tail/center ratio *grows* with budget rather than shrinking. The magnitude is largest on MMLU and smallest on LMentry, but the direction (worst prompt too low and/or best prompt too high) never reverses.

Table 3: The center–tail gap and over-dispersion generalize across benchmarks (PromptEval). "ratio" is tail/center mean-|error|; "o.d.%" is the fraction of cells whose estimated spread exceeds the truth (cells with true spread ≥ 0.02).

	B=200 (headline)				B=1600	
benchmark	q_{50} err	tail err	ratio	o.d.%	q_{50} err	ratio
MMLU (MCQ)	0.053	0.227	4.2×	100	0.006	10.1×
BBH (free-form)	0.070	0.198	2.8×	93	0.015	6.0×
LMentry	0.059	0.142	2.4×	85	0.022	3.4×

8 Related work

Prompt sensitivity of LLM benchmarks is well documented [Sclar et al., 2024, Mizrahi et al., 2024], motivating multi-prompt evaluation and PromptEval [Polo et al., 2024]. A parallel psychometric line applies item-response theory to LLM evaluation for efficiency and validity [Madaan et al., 2024]. Our work is a targeted reproduction and stress-test in the spirit of "sober-look" evaluations: we confirm the method's central-tendency claims and expose a specific, decision-relevant tail failure plus a remedy. Concurrent work questions tail-focused metrics in a different setting (toxicity reward-model extreme-value tails) [Anonymous, 2026b]; our over-dispersion mechanism and deconvolution correction for multi-prompt performance quantiles are, to our knowledge, unreported.

9 Limitations

Our study is a re-analysis of three released benchmarks with fixed template sets; the exact magnitudes are specific to those matrices and to the Rasch estimator. The correction is a variance (spread) correction and assumes the estimated *mean* is approximately unbiased, which holds empirically here. We analyze the released additive Rasch model; covariate-augmented variants may behave differently, though they share the additive-in-logits structure.

10 Conclusion

Multi-prompt evaluation delivers on its central promise but not on its tails: the very quantiles that make a distribution more informative than a point estimate are the least reliable, systematically over-dispersed, and not fixed by more budget. We give a diagnosis, a mechanism, and a label-free correction. The practical rule is simple: *trust the center, correct or distrust the edges.*

Reproducibility. All code, the exact-match reproduction, and figure-generating scripts run on a laptop from the authors' public data; see the repository.

References

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